DIGITAL TWIN PROTOTYPE

GlycoTwin AI

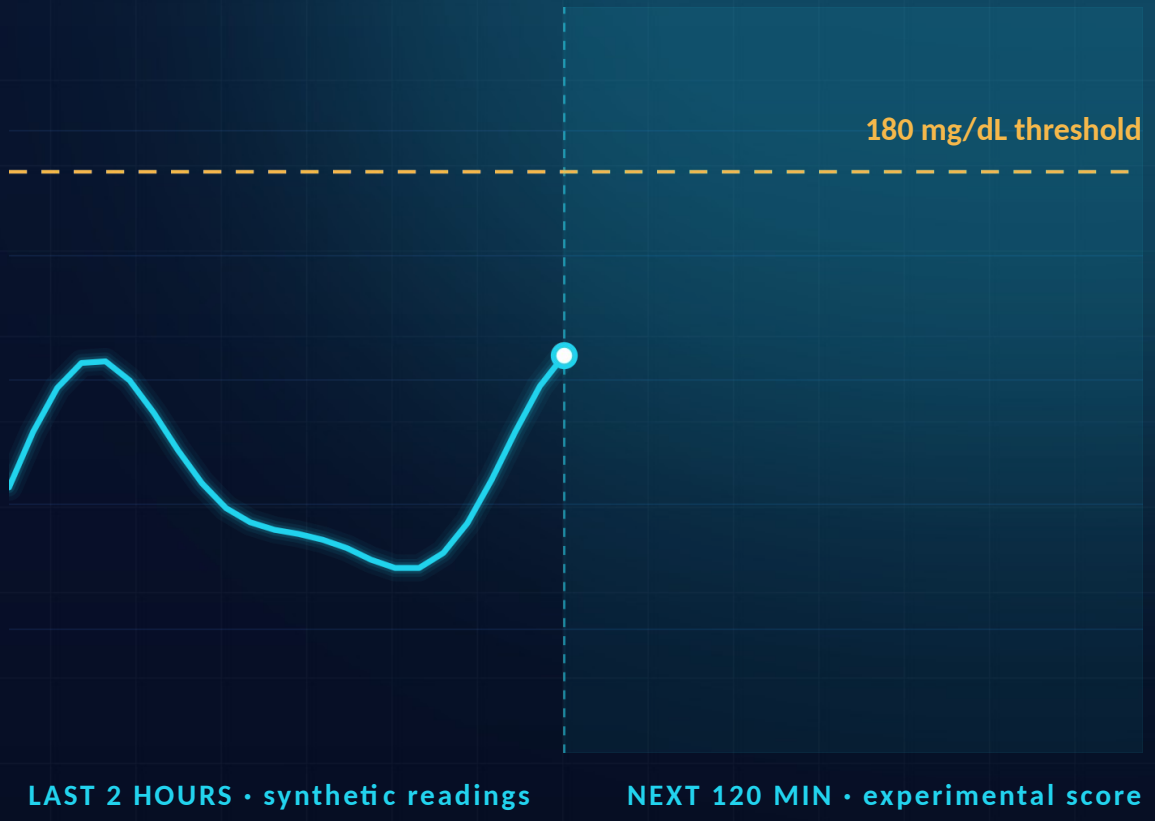
A transparent digital-twin prototype that turns glucose timelines into an inspectable two-hour excursion score.

Happiest Health Digital Twin Challenge 2026

[Team name] · [Institution]

SYNTHETIC DATA

EXPERIMENTAL PROTOTYPE



Curve: GT001 synthetic readings generated by the prototype API.

THE PROBLEM

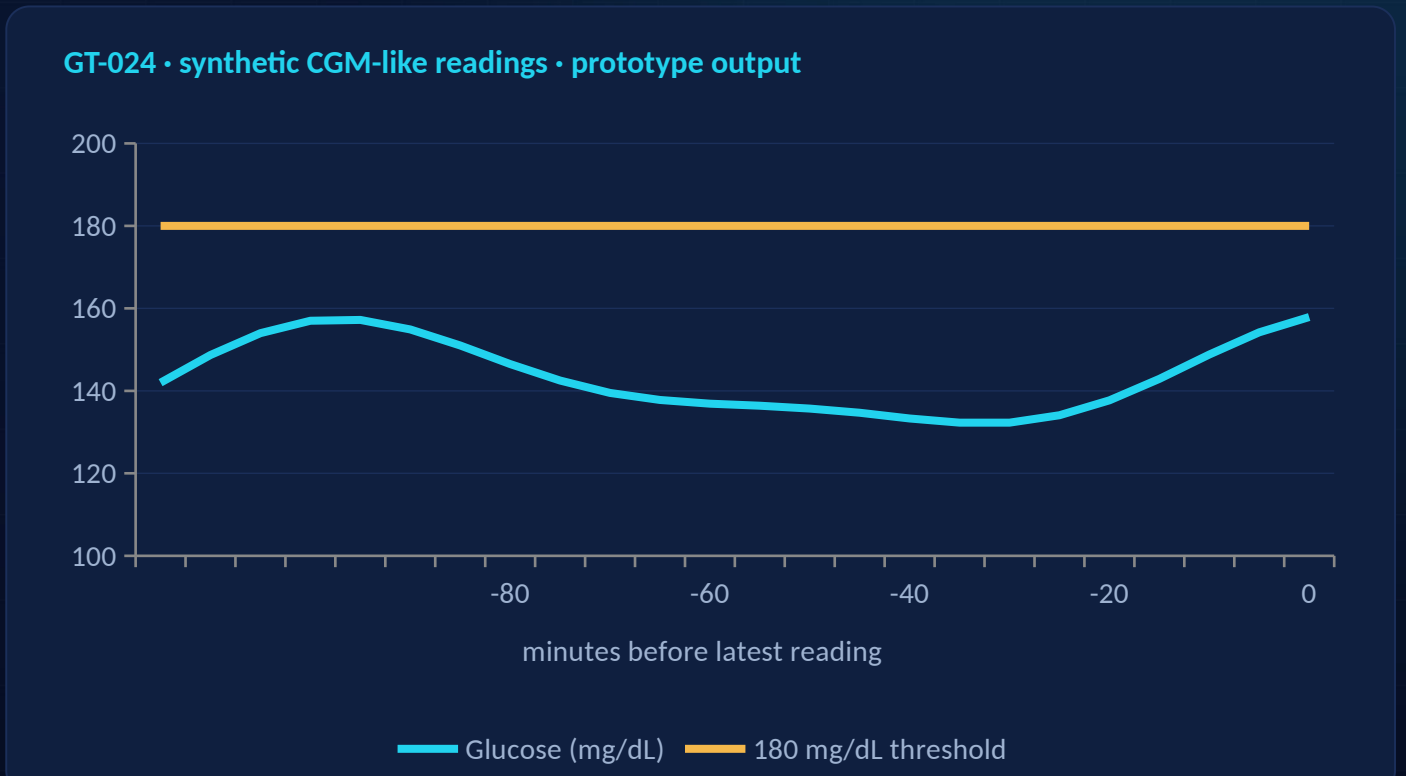
A single lab value can't show how glucose moves

A snapshot

HbA1c summarises average glucose over weeks. It does not show swings hour to hour.

A timeline

384 recent readings at 15-minute intervals show direction, swings and speed of change.



Our question: can recent readings give an inspectable signal of a likely rise above 180 mg/dL within two hours?

OUR APPROACH

A digital twin built from profile and timeline

Virtual patient profile

Age, diagnosis, years since diagnosis, HbA1c, baseline glucose

Glucose timeline

Timestamped readings every 15 minutes

Virtual patient twin

Profile and recent timeline held together as one patient state

Experimental 2-hour score

Probability of crossing 180 mg/dL

Historical replay

Step through earlier readings

Today's twin is a data model, not a physiological simulation.

USER JOURNEY

From virtual patient to prediction in five steps

01

Select patient

Choose from 36 synthetic virtual patients

`GET /api/patients`

02

Review profile

Age, diagnosis, HbA1c, baseline

`GET /api/patients`

03

View history

Recent readings from a 15-minute sensor stream

`GET …/timeline`

04

Inspect prediction

Score and experimental risk band

`GET …/predict`

05

Replay a point

Pick an earlier reading

`POST /api/simulate`

Example outputs vary by patient and current model run. Use the live dashboard for actual displayed values; all data are synthetic.

LIVE PROTOTYPE

The prototype, running end to end



Dashboard illustration · synthetic data

1. **Virtual patient selector**
   Switch between 36 synthetic patient profiles
2. **Prediction target**
   180 mg/dL within 120 minutes
3. **Safety banner**
   Synthetic, experimental, non-clinical
4. **Overview metrics**
   Glucose, probability, band, time
5. **Virtual patient profile**
   Age, diagnosis, HbA1c, baseline

IMPLEMENTED NOW

SYSTEM ARCHITECTURE

Three layers, one clear data path

PRESENTATION

Streamlit dashboard

- Patient selector
- Overview metrics
- Virtual patient profile
- Glucose timeline

HTTP · JSON →

SERVICE

FastAPI service

- `GET /health`
- `GET /api/patients`
- `GET /api/patient/{id}/timeline`
- `GET /api/patient/{id}/predict`
- `POST /api/simulate`

HTTP · JSON →

DATA & LOGIC

Synthetic data + trained ML

- 36 generated virtual patients
- CGM-like stream: 15-minute readings
- Random Forest model (experimental)

PLANNED — Not built yet: database · authentication · external clinical validation

SYNTHETIC-DATA MODEL

How the model estimates future risk

RANDOM FOREST · EXPERIMENTAL

Random Forest classifier
Uses past-only rolling glucose features
Fuses sensor history with EHR context

MODEL EVALUATION · HELD-OUT PATIENTS

Test set: 9 synthetic patients; 6,804 time points

ROC-AUC 0.9608 · Average precision 0.2258

Precision 47.6% · Recall 19.2% at threshold 0.5

RISK BANDS



Display bands are UI categories, not clinical classifications.

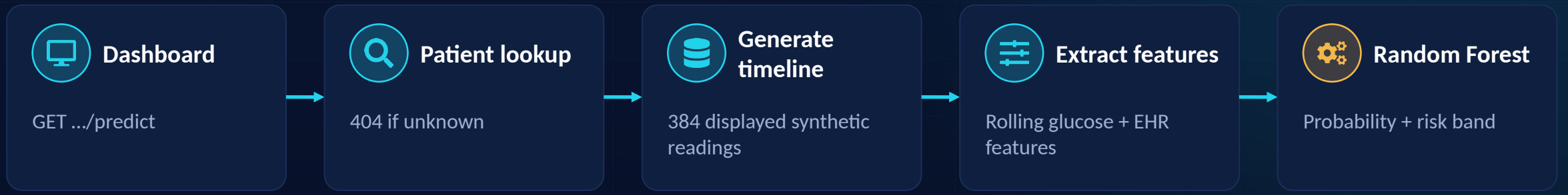
What it is not

Synthetic-data metrics do not establish clinical performance. Positive events are rare, so accuracy can mislead.

The model is experimental; compare it with a simple baseline and validate externally before clinical claims.

DATA AND API WORKFLOW

How a prediction request flows through the API



GET	/health	Service status
GET	/api/patients	Profiles
GET	/api/patient/{id}/timeline	Glucose history
GET	/api/patient/{id}/predict	Experimental probability
POST	/api/simulate	Historical replay

EXAMPLE RESPONSE · SYNTHETIC · SCHEMATIC

```
{
  "current_glucose_mg_dl": 157.9,
  "predicted_probability": 0.31,
  "risk_band": "LOWER",
  "threshold_mg_dl": 180,
  "horizon_minutes": 120,
  "model": "RandomForestClassifier"
}
Example values are illustrative; use live API
output.
```

FEASIBILITY AND IMPLEMENTATION

What works today, and what remains

IMPLEMENTED NOW

- Streamlit dashboard with 36 generated virtual patients
- FastAPI service with 5 endpoints
- Synthetic 15-minute glucose time series
- Architecture documentation

SYNTHETIC-DATA MODEL

- Random Forest 2-hour excursion estimate
- Experimental probability bands
- Target: 180 mg/dL within 120 min
- Synthetic metrics only; no clinical claim

PLANNED

- Validated, permitted datasets
- External validation and clinical review
- Clinician feedback
- Security and data governance

STACK

Python · FastAPI · Streamlit · Pandas · NumPy · Plotly

Scikit-learn Random Forest and Joblib power the prediction pipeline.

RESPONSIBLE HEALTHCARE AI

Built to be honest about what it isn't

Synthetic data only

No real patient data is used. Results say nothing about real people.

Not clinically validated

The model is trained on synthetic data; results are not clinically validated.

Privacy and governance

Real data would need consent, access control, security and permitted use.

Clinicians stay in charge

It does not diagnose, replace clinicians or guide medication doses.

Not for diagnosis · treatment · medication dosing · emergency monitoring

ROADMAP

A realistic path from prototype to evidence

0 — TODAY

Synthetic prototype

Dashboard, API, trained synthetic-data model

1 — DATA

Validated datasets

Permitted, de-identified data and governance

2 — MODEL

External validation

Independent permitted dataset; compare with baselines

3 — FEEDBACK

Clinician review

Usability and usefulness of outputs

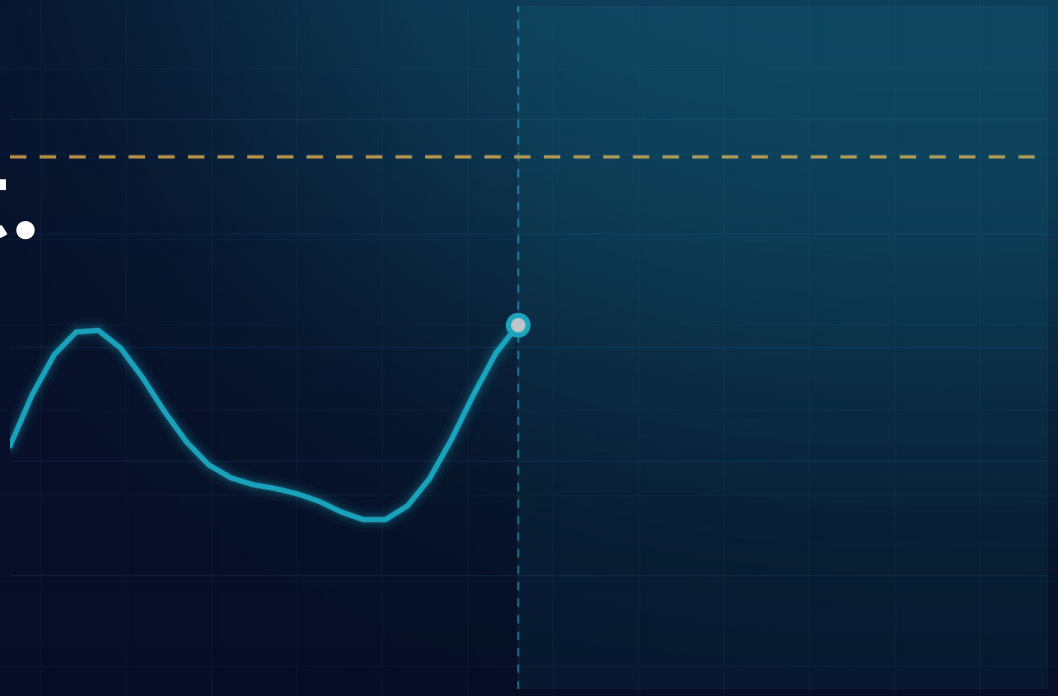
4 — PILOT

Monitored pilot

Only after validation and approvals

The current model remains experimental; external validation and clinical review are required before any real-world claims.

Honest about today.
Built for what comes next.



GlycoTwin AI combines synthetic EHR context and evolving CGM-like data to estimate a two-hour glucose excursion. External validation remains future work.

IMPLEMENTED EXPERIMENTAL PLANNED

[Team name] · [Institution] · [Contact]